FOREST FIRE PREDICTION

DATA ASSOCIATE 11 | GROUP 1

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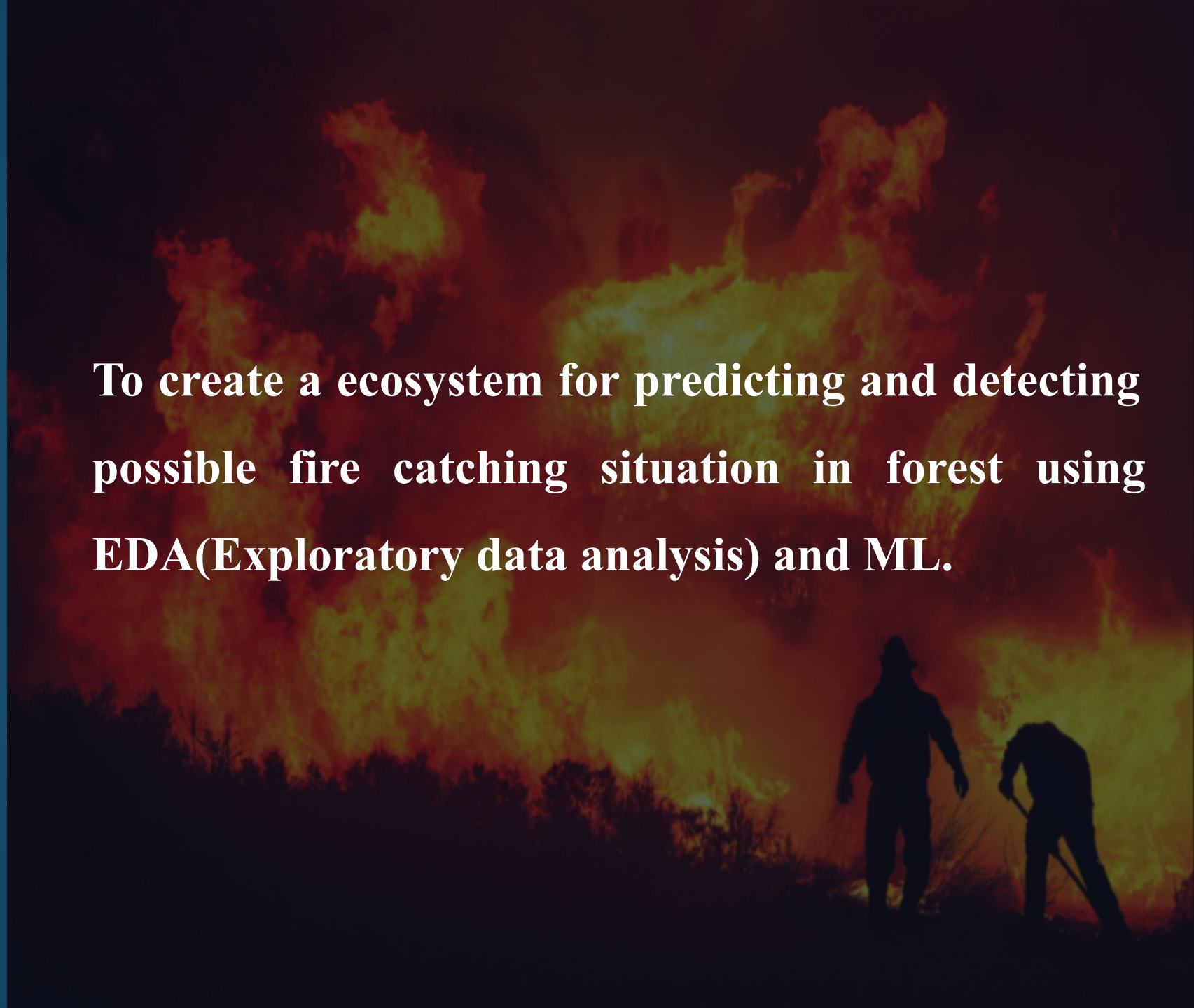
UNDER GUIDANCE OF MR. KUSHAL SHARMA SIR

BACKGROUND

- Major recurring phenomenon.
- Cost of damages more than $4 billion.
- Degradation of our environment.
- Increased frequency and intensity.
- Need regional indices for better assessment.
- Natural deforestation.
- Decreasing biodiversity.

OBJECTIVE

To create a ecosystem for predicting and detecting possible fire catching situation in forest using EDA(Exploratory data analysis) and ML.



PROBLEM-STATEMENT

The most common hazard in forests is forests fire. Forests fires are as old as the forests themselves. They pose a threat not only to the forest wealth but also to the entire regime to fauna and flora seriously disturbing the bio-diversity and the ecology and environment of a region. During summer, when there is no rain for months, the forests become littered with dry senescent leaves and twinges, which could burst into flames ignited by the slightest spark or by just a good combination of Heat-Fuel-Oxygen.

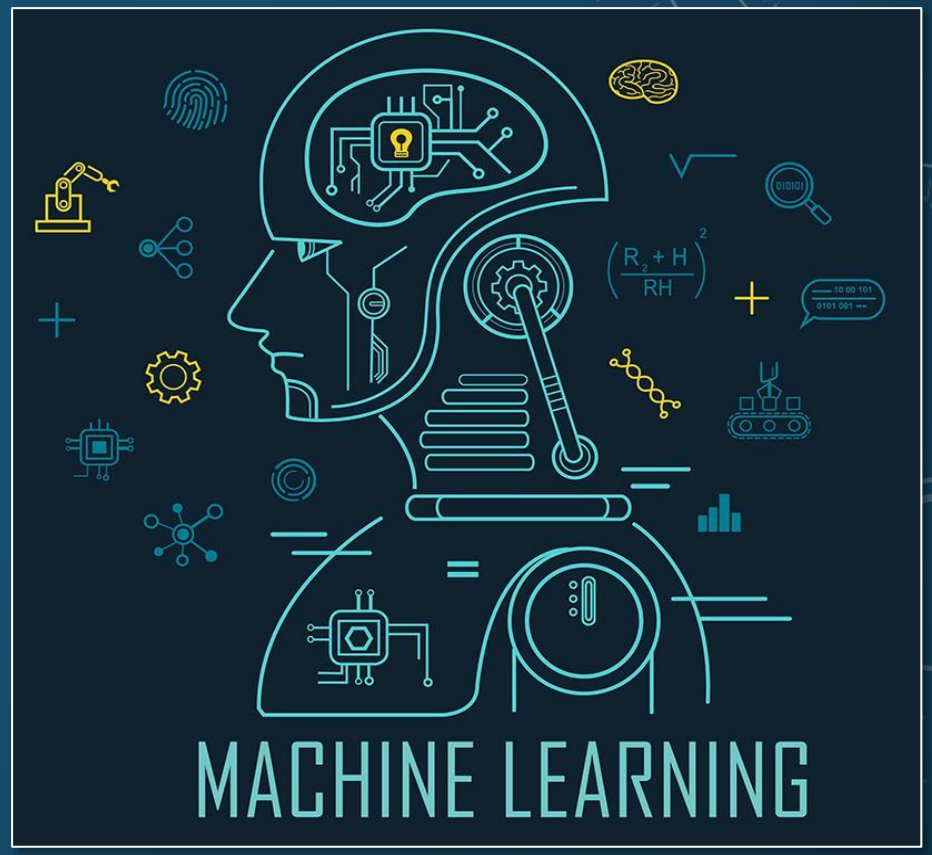
EXISTING SOLUTIONS

- Don’t set off pyrotechnics/ fireworks
- Carefully dispose of smoking material
- Camp responsibly
- Satellite Imaging

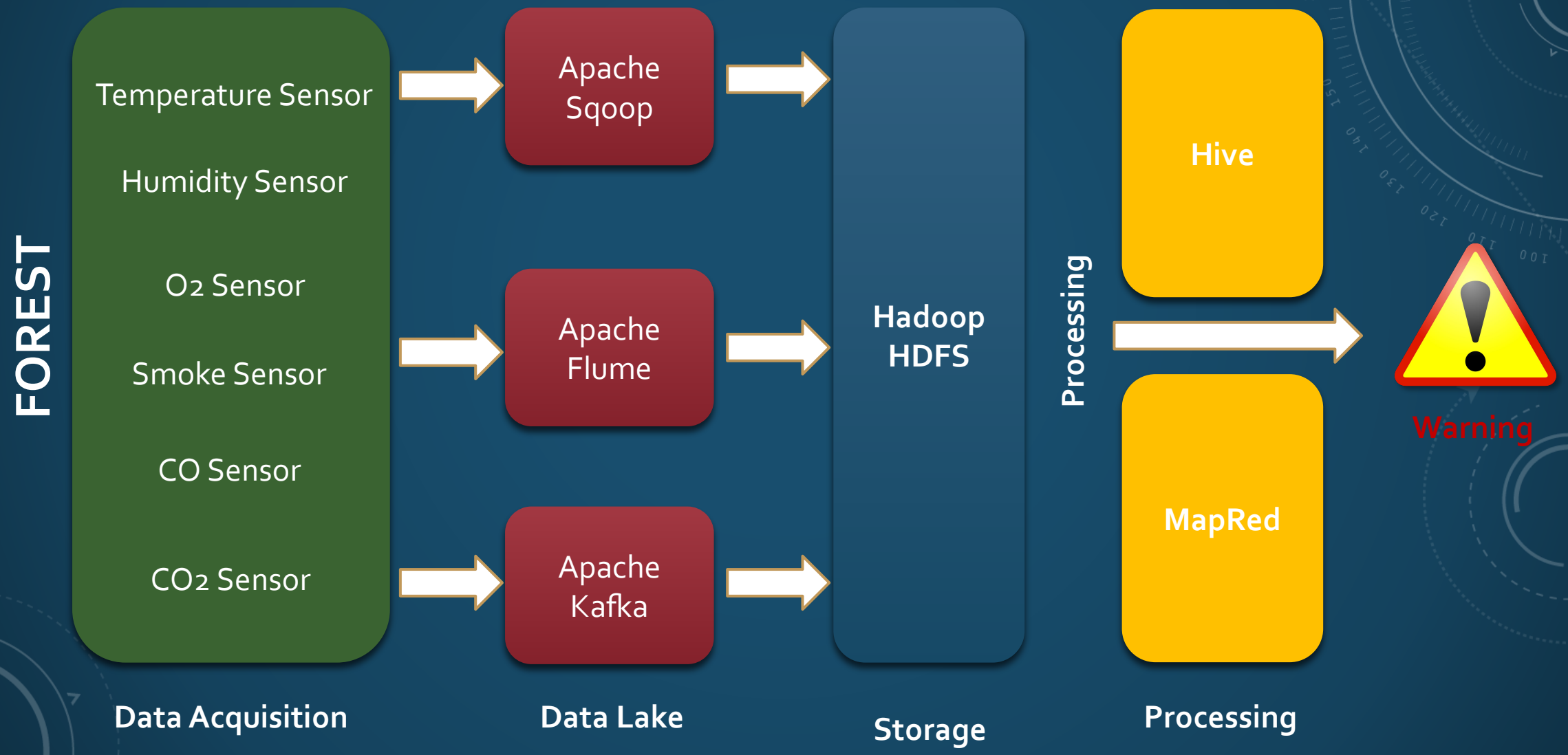
PROPOSED SOLUTION

Data Migration ,Exploratory data analysis on revived data along with machine learning techniques

Early detection and warning system.



ARCHITECTURE & ACTION DIAGRAM



TOOLS AND ALGORITHM

- Python Programming
- Hadoop | HDFS
- Apache Sqoop | Apache Flume | Apache Kafka (Data Migration)
- MySQL
- Tableau (Data Visualization)
- Pydoop

IMPLEMENTATION

1. Data Migration to HDFS using Apache Sqoop

Cloudera Quickstart 5.13.0...

Applications Places System Fri Oct 22, 8:51 AM cloudera

root@quickstart:/home/cloudera

File Edit View Search Terminal Help

```
mysql> insert into temperature_sensor values("11:58:15", 30.74);
Query OK, 1 row affected, 1 warning (0.00 sec)

mysql> insert into temperature_sensor values("13:18:15", 33.74);
Query OK, 1 row affected, 1 warning (0.00 sec)

mysql> select * from temperature_sensor;
+----------+---------+
| time     | temp    |
+----------+---------+
| 03:56:45 | 0.99999 |
| 04:46:35 | 0.99999 |
| 07:48:15 | 0.99999 |
| 11:58:15 | 0.99999 |
| 13:18:15 | 0.99999 |
+----------+---------+
5 rows in set (0.00 sec)

mysql> create table humidity_sensor(time TIME, humidity int, primary key (ti
;
Query OK, 0 rows affected (0.03 sec)

mysql> insert into humidity_sensor values("11:58:15", 47);
Query OK, 1 row affected (0.00 sec)

mysql> select * from humidity_sensor;
+----------+----------+
| time     | humidity |
+----------+----------+
| 11:58:15 |       47 |
+----------+----------+
1 row in set (0.00 sec)

mysql>
```

root@quickstart:/usr/lib/sqoop/lib

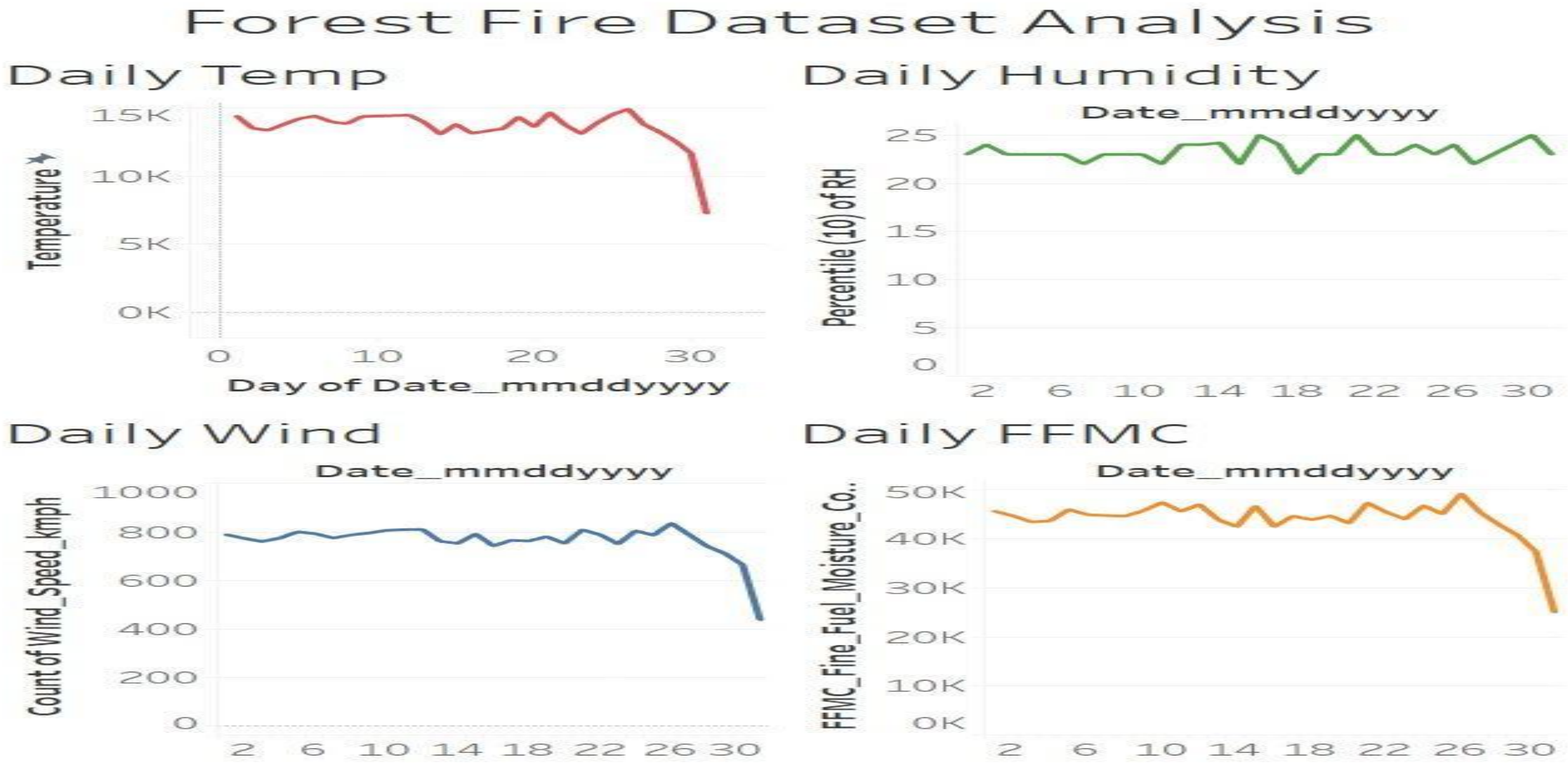
File Edit View Search Terminal Help

```
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=182
                CPU time spent (ms)=2100
                Physical memory (bytes) snapshot=134189056
                Virtual memory (bytes) snapshot=1510195200
                Total committed heap usage (bytes)=60882944
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=12
21/10/22 08:18:29 INFO mapreduce.ImportJobBase: Transferred 12 bytes
seconds (0.2275 bytes/sec)
21/10/22 08:18:29 INFO mapreduce.ImportJobBase: Retrieved 1 records.
[root@quickstart lib]# hdfs dfs -ls
Found 2 items
drwxr-xr-x   - root supergroup          0 2021-10-14 05:14 SqoopTest
drwxr-xr-x   - root supergroup          0 2021-10-22 08:18 humidity_s
[root@quickstart lib]# hdfs dfs -cat /user/root/humidity_sensor
cat: `/user/root/humidity_sensor': Is a directory
[root@quickstart lib]# hdfs dfs -ls /user/root/humidity_sensor
Found 2 items
-rw-r--r--   1 root supergroup          0 2021-10-22 08:18 /user/root
ensor/_SUCCESS
-rw-r--r--   1 root supergroup         12 2021-10-22 08:18 /user/root
ensor/part-m-00000
[root@quickstart lib]# hdfs dfs -cat /user/root/humidity_sensor/part-
11:58:15,47
[root@quickstart lib]#
```

[Browsing HDFS - Moz... root@quickstart:/usr/li... root@quickstart:/home...

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

DATA VISUALIZATION USING TABLEAU



OBSERVATION

- According to some performance parameters we found that the processing of MapRed works fine.
- We also have other options to carry out further in the project like processing with Apache Hive
- By changing some hyperparameters we managed to get precision as high as possible.

CONCLUSION

- We able to detect and predict forest fire before any disastrous event happens.
- Most of fires occurred during summer season .
- Quick action can be taken as the warning gets to the related personals.

LINKS

- Project_Process_01
- Project_Process_02
- Forest_Hive_Website
- Tableau_Analytics

THANK YOU.